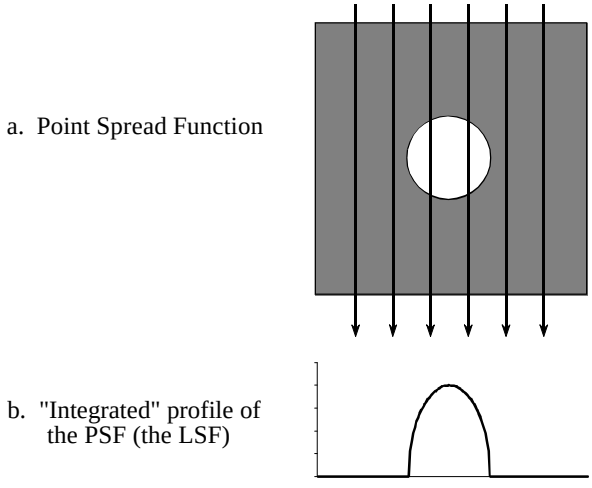


FIGURE 25-5
Relationship between the PSF and LSF. A pillbox PSF is shown in (a). Any row or column through the white center will be a rectangular pulse. Figure (b) shows the corresponding LSF, equivalent to an *integrated profile* of the PSF. That is, the LSF is found by sweeping across the image in some direction and adding (integrating) the pixel values along each ray. In the direction shown, this is done by adding all the pixels in each column.

has only one PSF, but an infinite number of LSFs, one for each angle. For example, imagine a system that has an oblong PSF. This makes the spatial resolution different in the vertical and horizontal directions, resulting in the LSF being different in these directions. Measuring the LSF at a single angle does not provide enough information to calculate the complete PSF except in the special instance where the PSF is circularly symmetric. Multiple LSF measurements at various angles make it possible to calculate a non-circular PSF; however, the mathematics is quite involved and usually not worth the effort. In fact, the problem of calculating the PSF from a number of LSF measurements is exactly the same problem faced in *computed tomography*, discussed later in this chapter.

As a practical matter, the LSF and the PSF are not dramatically different for most imaging systems, and it is very common to see one used as an approximation for the other. This is even more justifiable considering that there are two common cases where they are identical: the rectangular PSF has a rectangular LSF (with the same widths), and the Gaussian PSF has a Gaussian LSF (with the same standard deviations).

These concepts can be summarized into two skills: how to *evaluate* a resolution specification presented to you, and how to *measure* a resolution specification of your own. Suppose you come across an advertisement stating: "This system will resolve 40 line pairs per millimeter." You should interpret this to mean: "A sinusoid of 40 lp/mm will have its amplitude reduced to 3%-10% of its true value, and will be just barely visible in the image." You should also do the mental calculation that 40 lp/mm @ 10% contrast is equal to a 10%-90% edge response of 1/(40 lp/mm) = 0.025 mm. If the MTF specification is for a 3% contrast level, the edge response will be about 1.5 to 2 times wider.

When you measure the spatial resolution of an imaging system, the steps are carried out in reverse. Place a sharp edge in the image, and measure the